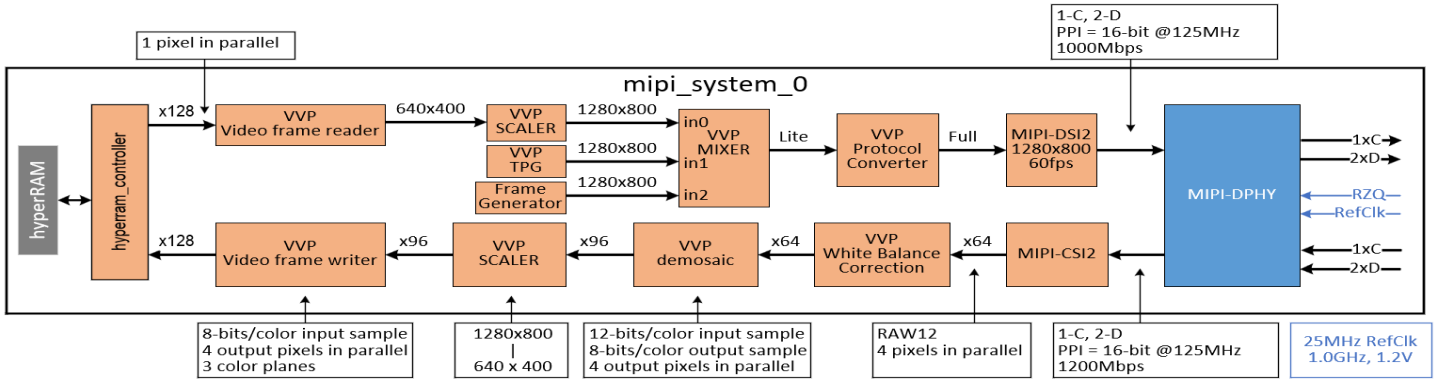


# Arrow FPGA Starter Kit – AXC3000 MIPI CSI to DSI Reference Design

**AXC3000 – Starter Kit**



This reference design showcases the MIPI capabilities of the Altera® Agilex™ 3 on the Arrow AXC3000. Live video from an ams OSRAM Mira220 or Sony IMX477 camera is buffered in HyperRAM and shown on a DSI LCD panel, with a touch overlay.
Camera data (RAW12 over two lanes) enters through the hard D-PHY and CSI-2 receiver, passes through white balance correction, demosaic and a scaler. Then it is stored by the Video Frame Writer in three rolling HyperRAM buffers. The Video Frame Reader then fetches each frame and scales it back up to 1280x800, a mixer adds the touch overlay, and the DSI-2 transmitter drives the panel at 1280x800 and 60fps.




A Nios® V/m soft processor initializes the camera and selects the video source through a simple terminal, delivering a complete camera-to-display vision pipeline that runs out of the box on the AXC3000.

**Key Features: Arrow AXC3000 Starter Kit**

- Agilex™ 3 - A3CY100BM16AE7S (M16A)
  - * 100 kLE
  - * 138 DSP blocks
  - * 276 multipliers 18x19
  - * 4.47 Mb internal memory
  - * MIPI D-PHY interface
  - * Secure Device Manager
- Triple-buffered video in HyperRAM
- OSRAM Mira220 or Sony IMX477 support
- Video & Vision Processing (VVP) pipeline
- Nios V/m soft processor

**Ordering number:** AXC3000
**Suggested Resale:** 129 USD
**Technical details:** GitHub page

**Partners:**